

1. A 16-year-old boy is brought to the office because of a 1-month history of intermittent sharp pain of the ring and small fingers of his right hand. He does not recall any trauma to the hand and has not had fever, weakness, chills, or rash. He has no history of serious illness and takes no medications. He is right-hand dominant. He plays varsity basketball and pitches for the varsity baseball team. He has a part-time after-school job bagging groceries at a local store. His mother has moderately severe rheumatoid arthritis. The patient's vital signs are within normal limits. Both wrists and hands appear normal. Muscle strength is 5/5 in the upper extremities. Range of motion of the shoulders, elbows, wrists, and finger joints is full bilaterally. Which of the following is the most likely cause of this patient's pain?

- ☐ A) Occupational tasks
- ☐ B) Osteoarthritis
- ☐ C) Sports activities
- ☐ D) Synovial crystal deposition
- ☐ E) The cause is idiopathic

2. Six hours after a 10-year-old boy undergoes operative repair of a humerus fracture, his parents notice that he is unresponsive and barely breathing and notify the nursing staff. The patient has no history of serious illness. His only medication is intravenous morphine. On evaluation, he is unconscious. His temperature is 37°C (98.6°F), pulse is 120/min, respirations are 10/min, and blood pressure is 90/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 88%. The pupils are 2 mm. The remainder of the examination shows no abnormalities. Intravenous naloxone is administered, and the patient regains consciousness. His respirations now are 18/min. It is most appropriate for the hospital to investigate this incident as which of the following?

- A) Act of negligence
- B) Improper hand-off
- C) Perioperative risk
- D) Quality indicator
- E) Sentinel event

3. An 82-year-old man comes to the physician because of an 8-month history of progressive urinary frequency and decreased urinary stream. He underwent transurethral resection of the prostate 11 years ago because of benign hyperplasia. He takes no medications. Examination shows a moderately enlarged prostate. His serum creatinine concentration has increased from 1.3 mg/dL 10 months ago to 2.4 mg/dL today. Which of the following is the **most appropriate initial step in management?**

- ☐ A) Measurement of postvoid residual volume
- ☐ B) Urine cytology
- ☐ C) Intravenous pyelography
- ☐ D) Retrograde urethrography
- ☐ E) CT scan of the abdomen

4. An 80-year-old woman comes to the office for a routine examination. She has osteoarthritis, hypertension, and peptic ulcer disease. Her medications are acetaminophen, lisinopril, and omeprazole. She says she has joint pain and stiffness, which limit her mobility and make it difficult for her to open her medication bottles, shop for groceries, cook, and clean. She lives alone. Vital signs are within normal limits. Examination shows bony enlargement of the shoulders, joints of the hands, and knees. There is crepitus. Range of motion is limited by joint tenderness. Which of the following is the most appropriate next step in management?

- ☐ A) Addition of a nonsteroidal anti-inflammatory drug to the medication regimen
- ☐ B) Intra-articular administration of a corticosteroid
- ☐ C) Multidisciplinary geriatrics assessment
- ☐ D) Orthopaedic consultation
- ☐ E) Placement in a nursing care facility

5. A 32-year-old man comes to the physician because of a 6-week history of mild headaches and fatigue. He has no history of serious illness and takes no medications. He does not smoke cigarettes. He drinks two to three beers nightly. Four months ago, he began a highly stressful new job, in which he has a verbally abusive boss. He says he has financial concerns and is worried about job security. He used to exercise three times weekly but stopped exercising 3 months ago because he is "too busy." He lives alone and is not in a romantic relationship. He appears tired and withdrawn and responds to questions with short phrases. Physical examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Alcohol use disorder
- ☐ B) Antisocial personality disorder
- ☐ C) Generalized anxiety disorder
- ☐ D) Major depressive disorder
- ☐ E) Post-traumatic stress disorder

6. A 55-year-old woman comes to the office as a new patient because she was dissatisfied with her last physician. Four months ago, she had the onset of moderate chest pain. Cardiac stress scintigraphy showed a small area of reversible ischemia. The patient was diagnosed with stable angina; after beginning metoprolol therapy, her chest pain decreased from twice to once weekly. She has no other history of serious illness and takes no other medications. Her temperature is 36.9°C (98.4°F), pulse is 62/min, respirations are 16/min, and blood pressure is 115/70 mm Hg. Examination shows no abnormalities. Her previous physician recommended that she undergo cardiac catheterization, but she declined after researching the procedure on the Internet. She says she has read that medication alone is an acceptable alternative to cardiac catheterization and that she is unwilling to accept the risks associated with the procedure. She agrees to add atorvastatin to her medication regimen. Which of the following is the most appropriate next step in management?

- ☐ A) Attempting to persuade the patient to undergo catheterization
- ☐ B) Referral to a cardiologist
- ☐ C) Repeat cardiac stress scintigraphy
- ☐ D) Stress echocardiography
- ☐ E) No further management is indicated at this time

7. A 19-year-old woman comes to the office because of a 1-month history of frequent nausea and feeling hot when others do not. During the past year, she has had a 4.5-kg (10-lb) weight gain despite no change in appetite; her weight is unchanged since her last visit 1 month ago. Menarche was at the age of 13 years. Since then, menses have occurred at irregular 28- to 42-day intervals with varying amount of flow. Her last menstrual period was 3 weeks ago and was "much lighter" than she expected. She is sexually active with one male partner; they use condoms inconsistently. She is 168 cm (5 ft 6 in) tall and weighs 77 kg (170 lb); BMI is 27 kg/m². Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is most likely to confirm the diagnosis?
- ☐ A) Determination of serum estrogen:progesterone ratio
 - ☐ B) Measurement of serum free thyroxine concentration
 - ☐ C) Measurement of serum prolactin concentration
 - ☐ D) Measurement of serum thyroid-stimulating hormone concentration
 - ☐ E) Measurement of urine β-hCG concentration



8. A 45-year-old man, who was diagnosed with type 2 diabetes mellitus 5 years ago, comes to the office for a routine follow-up visit. His diabetes mellitus has been fairly well controlled with metformin. At his last visit 3 months ago, his hemoglobin A_{1c} was 7.8%, and his metformin was increased from 850 mg twice daily to 1000 mg twice daily. He also takes glipizide and atorvastatin. Laboratory studies, including a complete metabolic panel, serum lipid studies, and screening of the urine for microalbumin done during the previous visit 3 months ago, were all within the reference ranges. In addition to measuring this patient's hemoglobin A_{1c}, which of the following is the most appropriate next step in evaluation at today's visit?

- ☐ A) Examination of the patient's feet
- ☐ B) Fasting serum glucose concentration
- ☐ C) Liver function tests
- ☐ D) Serum lipid studies
- ☐ E) Urine microalbumin concentration



Previous



Next



Lab Values



Calculator



Review



Help



Pause

9. A 15-month-old girl is brought to the physician by her mother because of a 5-day history of fever, difficulty sleeping, decreased appetite, and fussiness. She has no history of serious illness or known allergies. She receives no medications. She appears mildly ill but is interactive. She is at the 75th percentile for length and 50th percentile for weight. Her temperature is 38.3°C (101°F). Examination shows purulent nasal drainage. The left tympanic membrane is edematous and erythematous. The right tympanic membrane and oropharynx are normal. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Amoxicillin
- ☐ B) Azithromycin
- ☐ C) Cefuroxime
- ☐ D) Clindamycin
- ☐ E) Tetracycline

10. A 13-year-old boy is brought to the office by his mother for a well-child examination. He has no history of serious illness and takes no medications. Growth and development have been normal. His medical records indicate that vaccinations were up-to-date through age 9 years; he has not received any vaccinations since that time. Vital signs are within normal limits and physical examination shows no abnormalities. Administration of which of the following vaccines is most appropriate at this time?

- ☐ A) Hepatitis B
- ☐ B) Inactivated poliovirus
- ☐ C) Measles-mumps-rubella
- ☐ D) Meningococcal
- ☐ E) Pneumococcal
- ☐ F) Varicella

11. A 42-year-old man comes to the physician for a routine life insurance examination. He feels well. He has smoked one pack of cigarettes daily for 25 years and drinks two to four alcoholic beverages daily. His work as a journalist sometimes takes him to developing countries, but he has not traveled abroad during the past year. He has not had any immunizations during adulthood except for tetanus toxoid 12 years ago. He has never had a blood transfusion. Examination shows no abnormalities. Laboratory studies show:

Hemoglobin	16.2 g/dL
Serum	
Albumin	3.8 g/dL
Total bilirubin	1.1 mg/dL
Alkaline phosphatase	95 U/L
AST	68 U/L
ALT	44 U/L
Ferritin	50 ng/mL

Which of the following is most likely to have prevented these laboratory abnormalities?

- ☐ A) Abstinence from alcohol
- ☐ B) Administration of diphtheria-tetanus toxoid
- ☐ C) Administration of hepatitis A vaccine
- ☐ D) Low-protein, branched-chain amino acid-enriched diet
- ☐ E) Smoking cessation
- ☐ F) Weekly phlebotomy

12. An asymptomatic 27-year-old woman comes for a routine health maintenance examination. She works as a lawyer. Her temperature is 37.2°C (98.9°F), pulse is 96/min, respirations are 16/min, and blood pressure is 144/82 mm Hg. Serum studies show a thyroxine (T_4) concentration of 14 µg/dL, a triiodothyronine (T_3) concentration of 250 ng/dL, and a thyroid-stimulating hormone concentration of 0.1 µU/mL. A radioactive thyroid scan shows increased uptake and a diffusely enlarged gland. Which of the following is the most likely cause of this patient's findings?

- ☐ A) Acute suppurative thyroiditis
- ☐ B) Chronic lymphocytic thyroiditis (Hashimoto disease)
- ☐ C) Drug-induced hypothyroidism
- ☐ D) Euthyroid sick syndrome
- ☐ E) Exogenous administration of thyroid hormone
- ☐ F) Hyperthyroidism
- ☐ G) Multiple endocrine neoplasia syndrome
- ☐ H) Subacute thyroiditis
- ☐ I) Thyroid cancer

13. A 45-year-old man with type 2 diabetes mellitus and hypertension comes to the physician as a new patient. He has no other history of serious illness. He takes metformin and hydrochlorothiazide. He has smoked one pack of cigarettes daily for 20 years. His blood pressure is 142/82 mm Hg. The remainder of the examination shows no abnormalities. Laboratory studies from 1 month ago show:

Hemoglobin A _{1c}	7.8%
Serum	
Na ⁺	142 mEq/L
K ⁺	4.3 mEq/L
Cl ⁻	109 mEq/L
Urea nitrogen	14 mg/dL
Glucose	135 mg/dL
Creatinine	0.7 mg/dL
Cholesterol, total	214 mg/dL
HDL-cholesterol	50 mg/dL
LDL-cholesterol	120 mg/dL
Triglycerides	142 mg/dL

Which of the following modifications would be most beneficial to this patient?

- ☐ A) Smoking cessation
- ☐ B) Adding lisinopril therapy to the medication regimen
- ☐ C) Adding pravastatin therapy to the medication regimen
- ☐ D) Increasing the dosage of hydrochlorothiazide
- ☐ E) Increasing the dosage of metformin



14. A 72-year-old woman comes to the clinic because of a 6-year history of increasingly severe pain and progressively decreasing mobility of her left shoulder. She has congestive heart failure, coronary artery disease, kidney disease, and type 2 diabetes mellitus. Her medications are metoprolol, enalapril, and glipizide. Vital signs are within normal limits. Range of motion of the left shoulder is decreased and causes crepitus. Examination shows no other abnormalities. X-ray of the left shoulder is shown. Which of the following is the most appropriate next step in management?

- ☐ A) Celecoxib therapy
- ☐ B) Glucosamine supplementation
- ☐ C) Ibuprofen therapy
- ☐ D) Physical therapy
- ☐ E) Recommendation for an anti-inflammatory diet

15. A 17-year-old boy comes to the physician because of a 3-day history of increasingly severe pain of his right testicle. He also has had sweating, chills, and burning pain with urination during this time. He has no history of serious illness and takes no medications. His temperature is 38.5°C (101.3°F); other vital signs are within normal limits. Examination shows edema and tenderness to palpation behind the right testicle. Which of the following is the most appropriate next step in management?

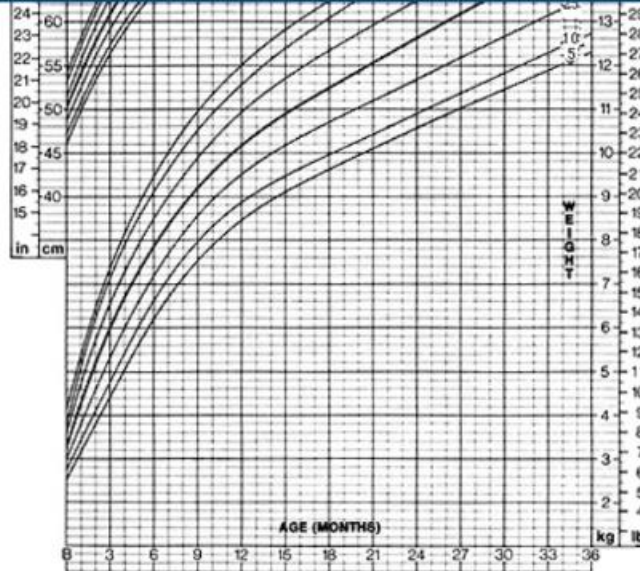
- ☐ A) Antibiotic therapy
- ☐ B) Scrotal elevation and warm baths
- ☐ C) Ultrasonography of the scrotum
- ☐ D) Referral to a urologist
- ☐ E) Bed rest and reassurance

16. A randomized, double-blind trial is performed to assess the efficacy of a new drug for the treatment of streptococcal pharyngitis. A total of 590 women between the ages of 20 and 34 years, who have culture-confirmed streptococcal pharyngitis, are enrolled; 295 women receive treatment with penicillin V potassium, and 295 receive treatment with the new drug. Results show bacteria eradication of 94.2% in the group treated with penicillin V potassium compared with 99.4% in the group treated with the new drug ($P=0.04$). Which of the following is the most accurate conclusion based on these findings?

- ☐ A) The new drug should be the gold standard for treatment of streptococcal pharyngitis
- ☐ B) The results are not statistically significant
- ☐ C) The study does not have a control group
- ☐ D) The study has good internal validity
- ☐ E) The study uses a case-control study design



17. A 64-year-old woman comes to the office because of a 1-year history of a nonhealing lesion on her left lower leg. She reports that the lesion does not itch, bleed, or drain. She has not attempted any treatments or remedies, and she does not wear any clothing or other items that rub the affected area. Vital signs are within normal limits. Examination of the left lower extremity discloses the findings shown in the photograph. The remainder of the physical examination shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Atypical melanoma
 - ☐ B) Basal cell carcinoma
 - ☐ C) Malignant melanoma
 - ☐ D) Psoriasis
 - ☐ E) Squamous cell carcinoma



18. A 4-month-old breast-fed male infant is brought to the clinic by his parents for a well-child examination. His birth weight was 3500 g (7 lb 11 oz) and his current weight is 7000 g (15 lb 7 oz). The parents ask if their child's weight gain is appropriate. The patient's growth and development are as shown in the growth chart. Vital signs are within normal limits and physical examination shows no abnormalities. Which of the following is the most appropriate statement to the parents about this patient's weight at this time?

- ☐ A) He should be switched from breast-feeding to bottle-feeding
- ☐ B) He should be taken to a nutritionist for further evaluation
- ☐ C) His thyroid-stimulating hormone concentration should be checked
- ☐ D) His weight gain is appropriate
- ☐ E) His weight gain is too rapid for his age

19. A 56-year-old postmenopausal woman with type 2 diabetes mellitus and hypertension comes to the clinic for a routine follow-up examination. Her parents immigrated from Libya. She says she has been feeling more tired than usual for the past 2 months; she has been sleeping well but becomes fatigued as the day progresses. She otherwise has no symptoms. Vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show:

Hemoglobin	10.5 g/dL
Hematocrit	33%
Mean corpuscular volume	75 μm^3
Red cell distribution width	17% (N=11.5–14.5)
Serum	
Ferritin	45 ng/mL
Thyroid-stimulating hormone	2.7 $\mu\text{U/mL}$

Which of the following is the most likely cause of this patient's current condition?

- ☐ A) Anemia of chronic disease
- ☐ B) Colon cancer
- ☐ C) Folic acid deficiency
- ☐ D) Iron deficiency anemia
- ☐ E) Thalassemia minor

20. A 50-year-old woman comes to the physician because of a 3-day history of swelling, redness, and aching pain from her right foot to just above her knee. There is no history of trauma. She has a 10-year history of hypertension and a 3-year history of type 2 diabetes mellitus. Her medications are metformin and hydrochlorothiazide. She works as a bus driver. Her temperature is 37.3°C (99.2°F), pulse is 88/min, respirations are 12/min, and blood pressure is 148/92 mm Hg. Examination shows an edematous, erythematous, tender right leg. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Complete blood count
- ☐ B) Measurement of serum hemoglobin A_{1c}
- ☐ C) Duplex ultrasonography of the right lower extremity
- ☐ D) Spiral CT scan of the chest

21. A 25-year-old woman who is a schoolteacher has a positive PPD skin test at a routine examination. She was exposed to a person with active tuberculosis when she was a child; however, all previous PPD skin tests since that time have been negative. Her last PPD skin test was 5 years ago. An x-ray of the chest shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Repeat PPD skin test in 10 weeks
- ☐ B) Repeat PPD skin test in 1 year
- ☐ C) Second strength PPD skin test (250 tuberculin units) now
- ☐ D) Treatment of latent tuberculosis now
- ☐ E) Treatment of active tuberculosis now
- ☐ F) No testing or therapy indicated

22. A 22-year-old woman comes to the office because of a 3-year history of episodes of abdominal discomfort and frequent diarrhea. During this time, she also has had a 4.5-kg (10-lb) weight loss. Two months ago, she decided to eliminate wheat products from her diet, which has improved her symptoms. She has no history of serious illness and takes no medications. She appears mildly anxious. She is 168 cm (5 ft 6 in) tall and weighs 52 kg (115 lb); BMI is 19 kg/m². Her pulse is 90/min, respirations are 16/min, and blood pressure is 102/58 mm Hg. Examination shows no other abnormalities. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Increased serum triiodothyronine (T₃) concentration
- ☐ B) Laxative abuse
- ☐ C) Transmural intestinal inflammation
- ☐ D) Ulceration of colonic mucosa
- ☐ E) Villous atrophy

23. An 81-year-old woman who recently emigrated from Ethiopia comes to the office because of a 1-year history of knee pain. The patient does not speak English and is accompanied by her son; a medical interpreter is used. Two weeks ago, she was evaluated by another provider in the practice. Results of laboratory screening studies ordered at that time were positive for syphilis, tuberculosis, and hepatitis C. CT scan of the liver obtained 1 week ago was suspicious for late-stage hepatocellular carcinoma. A biopsy is scheduled for next week. The patient's son asks to speak with the physician alone. He says that he has not told the patient about the possible cancer diagnosis for cultural reasons. Which of the following is the most appropriate next step in management?

- ☐ A) Ask the patient how much she desires to know about her medical condition
- ☐ B) Cancel the biopsy appointment
- ☐ C) Do not discuss the results with the patient or her son to respect their cultural practices
- ☐ D) Honor the stated cultural wishes of the son and speak to him instead of the patient
- ☐ E) Treat the patient's latent syphilis

24. An asymptomatic 37-year-old man with alcohol use disorder comes for an initial examination. He drinks approximately 12 cans of beer daily. Examination shows palmar erythema and multiple spider angiomas over the upper back and chest. The spleen is palpated 3 cm below the left costal margin. Abdominal examination shows shifting dullness. Despite the physician's strong recommendation for the patient to stop drinking alcohol, he says he has no intention of abstaining. Which of the following is the most appropriate immunization for this patient?

- ☐ A) Acellular pertussis
- ☐ B) *Haemophilus influenzae* type b, conjugated
- ☐ C) Live *Salmonella typhi*
- ☐ D) Meningococcal
- ☐ E) 23-Valent pneumococcal

The following vignette applies to the next 2 items. The items in the set must be answered in sequential order. Once you click **Proceed to Next Item**, you will not be able to add or change an answer.

A 66-year-old woman comes to the office for a routine health maintenance examination. Her last menstrual period was 10 years ago. She has no history of serious illness and takes no medications. Five years ago, a routine Pap smear showed no abnormalities and human papillomavirus testing was negative. A mammogram obtained 1 year ago showed no abnormalities. A colonoscopy done 8 years ago showed no abnormalities. She does not smoke cigarettes. She has been in a monogamous relationship for the past 40 years. Vital signs are within normal limits and physical examination shows no abnormalities.

25. **Item 1 of 2**
Which of the following screening tests is most appropriate at this time?
- ☐ A) DEXA scan
 - ☐ B) Measurement of serum CA-125 concentration
 - ☐ C) Pap smear
 - ☐ D) Test of the stool for occult blood
 - ☐ E) Ultrasonography of the abdominal aorta

Proceed to Next Item

You must select "Proceed to Next Item" to view the next item.

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26. **Item 2 of 2**

The patient returns to the office 2 weeks later to discuss the results of her screening tests. She has started calcium and vitamin D supplementation. Results of the DEXA scan show a lumbar spine T-score of -2.8. Serum lipid studies show a total cholesterol concentration of 230 mg/dL and HDL-cholesterol concentration of 60 mg/dL. Her estimated 10-year arteriosclerotic cardiovascular disease risk is 5.7%. Which of the following is the most appropriate next step in treatment?

- ☐ A) Alendronate therapy
- ☐ B) Atorvastatin therapy
- ☐ C) Calcitonin therapy
- ☐ D) Omega-3 fish oil supplementation
- ☐ E) Raloxifene therapy
- ☐ F) No further treatment is indicated at this time

27. A 62-year-old man comes to the physician because of a 2-month history of shortness of breath with exertion. He also has had paroxysmal nocturnal dyspnea and orthopnea. He has type 2 diabetes mellitus and coronary artery disease; he had a myocardial infarction 3 years ago. Current medications include simvastatin and aspirin. His pulse is 100/min, respirations are 12/min, and blood pressure is 100/60 mm Hg. Crackles are heard on auscultation. There is an S₃. An x-ray of the chest shows cardiomegaly, cephalization of blood flow, and Kerley B lines. Which of the following is the most appropriate pharmacotherapy to improve this patient's chances of survival?

- ☐ A) Angiotensin-converting enzyme (ACE) inhibitor
- ☐ B) Calcium-channel blocking agent
- ☐ C) Digitalis
- ☐ D) Diuretic
- ☐ E) Nitrates

28. A 65-year-old man comes to the office because of a 5-year history of slowly worsening constipation. During this time, his stools have been hard and have decreased in frequency. He reports often having to strain and a sensation of incomplete evacuation following bowel movements. This is the first time he has sought treatment. Colonoscopy 5 years ago showed no abnormalities. He has a history of hypertension for which he takes hydrochlorothiazide. Vital signs are within normal limits. Physical examination discloses a palpable colon with hard stool over the left of the abdomen; the remainder of the physical examination shows no other abnormalities. Results of laboratory studies are within the reference ranges. In addition to increasing fluid intake, which of the following is the most appropriate initial pharmacotherapy?

- ☐ A) Lactulose
- ☐ B) Linaclotide
- ☐ C) Lubiprostone
- ☐ D) Magnesium citrate
- ☐ E) Psyllium

29. A 65-year-old man comes to the physician to request prostate cancer screening. He has not had urinary symptoms. He has no history of serious illness and takes no medications. His father had prostate cancer at the age of 85 years. Examination of the patient shows no abnormalities. Which of the following is the most appropriate next step regarding prostate cancer screening for this patient?

- ☐ A) Digital rectal examination and measurement of serum prostate-specific antigen (PSA) concentration
- ☐ B) Digital rectal examination only
- ☐ C) Discussing the risks and benefits of prostate cancer screening options and allowing the patient to decide
- ☐ D) Measurement of serum PSA concentration only
- ☐ E) Prostate cancer screening is not indicated for this patient

30. A 55-year-old man comes to the office for a routine examination. Eighteen months ago, he had progressive difficulty reading and "seeing small things." His symptoms initially improved with the use of weak reading glasses; he then required stronger reading glasses. He reports no other current problems with his vision and is otherwise asymptomatic. He has hypertension treated with hydrochlorothiazide. Vital signs are within normal limits. Visual acuity is 20/30 in the left eye and 20/20 in the right eye. No other abnormalities are noted. Which of the following is the most appropriate next step in management?
- ☐ A) Glaucoma screening
 - ☐ B) Measurement of fasting serum glucose concentration
 - ☐ C) Referral to an ophthalmologist
 - ☐ D) Ultrasonography of the carotid arteries
 - ☐ E) No further management is indicated

31. A 72-year-old man has had increasing shortness of breath for 6 months. He has lost 1.4 kg (3 lb) in the last 3 months, but his appetite has not decreased. He has had a nonproductive cough for more than 40 years. He has smoked two packs of cigarettes daily for 50 years and worked from age 30 to 40 years in a factory making automobile brake linings. He is afebrile and not cyanotic. There is dullness to percussion; on auscultation, diminished breath sounds and egophony over the right lower chest are noted. X-rays of the chest show a right pleural effusion and a mass contiguous with the pleura in the right lateral chest. There are diffuse irregular linear opacities that are most dense in the lower lung fields. Which of the following is the most likely diagnosis?

- ☐ A) Anthracosis
- ☐ B) Asbestosis
- ☐ C) Berylliosis
- ☐ D) Byssinosis
- ☐ E) Silicosis

32. A 52-year-old Egyptian woman comes to the office with her adult daughter because of a 5-day history of headaches. The patient is visiting her daughter in the United States for 3 months and does not speak much English. The daughter offers to translate for her mother and reports that the patient has a history of hypertension. She says that the patient's medications are amlodipine, lisinopril, and hydrochlorothiazide, and further explains that the patient ran out of her medications 2 weeks ago. Blood pressure is 175/105 mm Hg; remaining vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Arrange for telephone interpreter services
 - ☐ B) Ask the daughter to step out of the room so the patient can be fully examined
 - ☐ C) Obtain an ECG
 - ☐ D) Refill the patient's prescriptions and schedule a follow-up appointment in 2 days
 - ☐ E) Send the patient to the emergency department for further evaluation

33. A 35-year-old man comes to the office because of a 3-month history of fatigue, increased sleepiness, and feeling that he cannot enjoy his usual activities. During this time, he also has had decreased appetite, resulting in a 4.5-kg (10-lb) weight loss. He has no history of serious illness and takes no prescribed medications. Three weeks ago, he was fired from his job because of decreased productivity. He describes his marriage as "good" but says there have been recent marital strains because of financial concerns. Which of the following is the most appropriate additional history to obtain from this patient?

- ☐ A) Frequency of the couple's sexual activity during the past month
- ☐ B) The patient's alcohol consumption
- ☐ C) Whether the patient has a financial advisor
- ☐ D) Whether the patient has sought other means of employment
- ☐ E) Whether the patient takes over-the-counter medications

34. A 48-year-old woman with mitral valve prolapse and mild mitral regurgitation comes to the physician for a follow-up examination. At her last visit 12 weeks ago, her pulse was 76/min, and blood pressure was 138/82 mm Hg. She says she feels well. Eight months ago, she began a low-sodium diet and exercise program to decrease her blood pressure. She takes no medications. She is allergic to penicillin. Her pulse is 72/min, and blood pressure is 124/76 mm Hg. On cardiac examination, a grade 2/6 systolic murmur is heard best at the apex. The remainder of the examination shows no abnormalities. In 1 week, she will undergo a dental cleaning. Which of the following is the most appropriate prophylaxis for this patient?

- ☐ A) Amoxicillin
- ☐ B) Cefazolin
- ☐ C) Clarithromycin
- ☐ D) Clindamycin
- ☐ E) No prophylaxis is required

35. A 24-year-old woman, gravida 1, para 1, comes to the office because of a 1-day history of moderate right lower abdominal pain and vaginal bleeding. One year ago, her uncomplicated pregnancy ended in a spontaneous vaginal delivery of a healthy newborn at term. Her last menstrual period was 7 weeks ago; previously, menses occurred at regular 28-day intervals. Five years ago, she had pelvic inflammatory disease, which resolved with cefoxitin and doxycycline therapy. She has smoked one pack of cigarettes daily for 7 years; 4 days ago, she began varenicline therapy. She takes no other medications. Prior to her pregnancy, she used a copper IUD; she currently uses no contraception. She is 163 cm (5 ft 4 in) tall and weighs 66 kg (145 lb); BMI is 25 kg/m². Her temperature is 37.2°C (99.0°F), pulse is 88/min, respirations are 16/min, and blood pressure is 114/76 mm Hg. Abdominal examination shows moderate tenderness to palpation of the right lower quadrant; there are no palpable masses. Pelvic examination shows a closed cervical os; there is no blood or discharge. There is mild cervical motion tenderness. No adnexal masses are palpated. A urine pregnancy test is positive. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Culdocentesis
- ☐ B) Laparoscopy
- ☐ C) Measurement of serum β -hCG concentration
- ☐ D) Testing for *Chlamydia trachomatis* and *Neisseria gonorrhoeae*
- ☐ E) Transvaginal ultrasonography

36. A 36-year-old man is brought to the office by his running partner after he completed a 26.2-mile marathon. Shortly after the race, the patient reported nausea, moderate headache, and dizziness. He also had difficulty maintaining his balance and was oriented to person but not place or time. His running partner reports that the patient drank "a lot of fluids" during the race because the ambient temperature during the race was 94.0°F. The patient has no history of serious illness and takes no medications. He appears pale and is diaphoretic. He is oriented to day and time but not to place. His temperature is 37.0°C (98.6°F), pulse is 90/min, respirations are 16/min, and blood pressure is 144/84 mm Hg. Examination shows no other abnormalities. Which of the following laboratory findings is most likely to be noted in this patient?

- ☐ A) Hyperkalemia
- ☐ B) Hypernatremia
- ☐ C) Hypoglycemia
- ☐ D) Hypokalemia
- ☐ E) Hyponatremia

37. A 46-year-old man with thrombocytopenia comes to the office for a preoperative evaluation prior to the extraction of four wisdom teeth. The patient says he feels well. He takes no medications. Vital signs are within normal limits and physical examination shows no abnormalities. A complete blood count discloses a platelet count of $110,000/\text{mm}^3$. Which of the following is the most appropriate recommendation for this patient?

- ☐ A) Begin desmopressin therapy
- ☐ B) Cancel the dental extraction
- ☐ C) Proceed with dental extraction
- ☐ D) Reschedule another appointment to establish platelet count prior to the dental extraction
- ☐ E) Transfuse platelets prior to dental extraction

38. A 26-year-old woman comes to the office because of a 12-hour history of moderate chest pain and cough productive of blood-tinged sputum. She has no other history of serious illness. Her only medication is an oral contraceptive. She took a 10-hour airplane flight 2 days ago. She appears to be in mild respiratory distress. Her temperature is 37.3°C (99.1°F), pulse is 110/min and regular, respirations are 18/min, and blood pressure is 132/94 mm Hg. There is no cyanosis. Cardiopulmonary examination shows no abnormalities. There is erythema and edema of the right calf. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) CT angiography of the chest
- ☐ B) Determination of mixed venous oxygen saturation
- ☐ C) Serum D-dimer assay
- ☐ D) Venography of the right lower extremity
- ☐ E) Venous duplex ultrasonography of the right lower extremity

39. A 15-year-old boy is brought to the physician by his mother for examination prior to participation in school sports. He has no history of serious illness and takes no medications. There is a strong family history of obesity, and his mother struggles to keep her weight in the normal range. His father, maternal aunt, and maternal grandmother have type 2 diabetes mellitus. The patient says he eats healthy food, but on questioning, he reports that yesterday he ate 12 oz of frosted cereal with whole milk for breakfast; a hot dog, potato chips, cookie, fruit cocktail, and soda for lunch; a candy bar for an after-school snack; and a large fast-food cheeseburger, extra large French fries, and a chocolate milkshake for dinner. He watches television and plays video games when he is not in school. He is popular and socially active, but he does not have a girlfriend. He is obese but otherwise appears healthy. He is 160 cm (5 ft 3 in) tall and weighs 90 kg (198 lb); BMI is 35 kg/m². His temperature is 37.1°C (98.8°F), pulse is 87/min, respirations are 22/min, and blood pressure is 128/90 mm Hg. Examination shows moderate acne over the face. Genital and pubic hair development are sexual maturity rating stage 4. His hemoglobin A_{1c} is 6.3%, and his fingerstick blood glucose concentration is 128 mg/dL. Which of the following is the most appropriate next step in management?

- ☐ A) Counseling the patient on the importance of aerobic exercise
- ☐ B) Recommending an intensive weight-lifting regimen
- ☐ C) ACE inhibitor therapy
- ☐ D) Combination insulin therapy
- ☐ E) Hypoglycemic agent therapy

40. A 72-year-old woman is brought to the physician by her son because of a 5-day history of mild left arm pain. She reports that she "must have bumped the wall" when she was getting off the toilet. She has mild dementia, Alzheimer type, and rate-controlled atrial fibrillation. Her only medication is warfarin. She lives with her son. Her two other children live in the same city as the patient and visit her regularly. The patient's pulse is 84/min and irregularly irregular. Physical examination shows several ecchymoses over the upper extremities and older, healing ecchymoses over the forearms. On mental status examination, she has a slightly depressed mood and normal affect. She recalls 0 of 3 items after 5 minutes and is oriented to self and place only. X-rays of the upper extremities show no abnormalities. On further questioning, the patient's son says he does not know how she got the bruises. Without her son present, the patient is asked if anyone has hurt her; she appears uncomfortable and says she is clumsy. Which of the following is the most appropriate next step in management?
- ☐ A) Call the patient's other children to inquire about the injuries
 - ☐ B) Call the police and have them come to the office immediately
 - ☐ C) Contact senior protective services
 - ☐ D) Document the physical findings in the patient's medical record and ask the patient again about the injuries at her next routine examination
 - ☐ E) Order a complete skeletal series

41. A 5-year-old boy is brought to the office by his mother for a well-child examination. The patient's mother reports that he has been more fidgety lately and has not been able to focus on his school work for more than a few minutes at a time. He also does not multitask well. The patient is otherwise healthy, sleeps well, and has a happy demeanor. He has no history of serious illness and receives no medications. Vaccinations are up-to-date. Vital signs are within normal limits, and physical examination discloses no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Ask parents and teachers to complete validated behavioral rating scales
- ☐ B) Initiate methylphenidate therapy
- ☐ C) Obtain genetic testing
- ☐ D) Recommend decreasing dietary sugar intake
- ☐ E) Reassure the patient's mother this is normal behavior at this age

42. Over the past 4 days, an 8-year-old girl has had joint pain, first in her right knee and then in her left ankle. She had a severe sore throat 2 weeks ago after a camping trip in Georgia. Her temperature is 38.9°C (102°F), pulse is 128/min, and blood pressure is 108/70 mm Hg. The right knee and left ankle are red, swollen, and exquisitely tender. The lungs are clear to auscultation. A grade 3/6, holosystolic apical murmur is heard. Which of the following is the most likely diagnosis?

- ☐ A) Acute rheumatic fever
- ☐ B) Juvenile idiopathic arthritis
- ☐ C) Lyme disease
- ☐ D) Rocky Mountain spotted fever
- ☐ E) Serum sickness



43. A 22-year-old man comes to the physician because of a 3-month history of thick, flaky fingernails. During this time, he also has had dry, scaly, red patches of skin on his forearms and chest. He has no history of serious illness and takes no medications. During the past 2 years, he has worked as a restaurant dishwasher. A photograph of the left hand and right forearm is shown; similar findings are seen over the left forearm and trunk. Which of the following is the most **likely diagnosis**?

- A) Atopic dermatitis
- B) Hydradenitis suppurativa
- C) Nummular eczema
- D) Psoriasis
- E) Tinea corporis



44. An 11-year-old boy is brought to the physician because of a 2-day history of nasal congestion and mild sore throat; he has coughing and sneezing productive of green mucus. He has not had fever or ear pain. He has had no sick contacts. His temperature is 37°C (98.6°F), and pulse is 84/min and regular. Examination shows clear and mobile tympanic membranes. The tonsils are mildly enlarged and erythematous; there is no exudate. There is postnasal drainage in the posterior pharynx and copious yellow-green nasal drainage. The lungs are clear to auscultation. Heart sounds are normal. Which of the following is the most appropriate next step in management?

- ☐ A) Provide symptomatic care only
- ☐ B) Recommend sinus lavage in the clinic
- ☐ C) Culture of the nasal discharge
- ☐ D) X-ray of the sinuses
- ☐ E) Antiviral therapy
- ☐ F) Broad-spectrum antibiotic therapy

45. A 77-year-old man with coronary artery disease comes to the physician for a follow-up examination. He underwent coronary artery bypass grafting 4 years ago. He has a 3-year history of intermittent claudication. His blood pressure measurements from his last eight visits show:

Number of Months Ago	Blood Pressure (mm Hg)
24	130/80
21	130/60
15	140/70
12	130/80
8	160/90
6	170/96
4	180/110
2	180/100

Current medications include aspirin and atorvastatin. His temperature is 36.5°C (97.7°F), pulse is 80/min, and blood pressure is 180/110 mm Hg. Heart sounds are normal. Dorsalis pedis and posterior tibial pulses are not palpable bilaterally. Serum studies show:

K ⁺	3.1 mEq/L
HCO ₃ ⁻	28 mEq/L
Urea nitrogen	20 mg/dL
Glucose	90 mg/dL
Creatinine	1.1 mg/dL

Which of the following is the most likely cause of this patient's increased blood pressure?

- ☐ A) Dissecting abdominal aortic aneurysm
- ☐ B) Essential hypertension
- ☐ C) Renal artery stenosis
- ☐ D) Renal failure
- ☐ E) White-coat hypertension



46. A 76-year-old man comes to the physician because of a 5-year history of progressive redness and scaling over bald areas of his scalp. He has hypertension, hypercholesterolemia, and psoriasis over his arms. His medications are lisinopril, hydrochlorothiazide, lovastatin, clobetasol, and aspirin. He is a retired high school sports coach. Vital signs are within normal limits. A photograph of the scalp is shown. There are large areas of scaly, dry rash over the upper extremities. Which of the following is the most likely diagnosis of this patient's scalp findings?

- ☐ A) Actinic keratosis
- ☐ B) Basal cell carcinoma
- ☐ C) Melanoma
- ☐ D) Psoriasis
- ☐ E) Seborrheic dermatitis

47. A 56-year-old man comes to the office because of a 1-month history of a nonhealing lesion on his tongue. He says he thinks the lesion may have appeared when he bit his tongue while sleeping. He has a 2-year history of type 2 diabetes mellitus well controlled with metformin and diet. He works as an electrician at a shipyard. He has smoked one pack of cigarettes daily for 17 years. He drinks three 12-oz beers weekly. Vital signs are within normal limits. There is a 2-cm, ulcerated lesion over the left aspect of the tongue. The remainder of the examination, including that of the head and neck, shows no abnormalities. Which of the following is most likely to have prevented this patient's lesion?

- ☐ A) Alcohol abstinence
- ☐ B) Human papillomavirus vaccination
- ☐ C) Smoking cessation
- ☐ D) Use of a mask at work
- ☐ E) Use of a mouth guard at night



Previous



Next



Lab Values



Calculator



Review



Help



Pause

48. A 37-year-old woman comes to the physician because of a 3-month history of frequent headaches. She has no history of serious illness and takes no medications. Her last health maintenance examination 3 years ago showed no abnormalities. There is no family history of serious illness. She is 165 cm (5 ft 5 in) tall and weighs 66 kg (145 lb); BMI is 24 kg/m². Her temperature is 37.1°C (98.8°F), pulse is 88/min, respirations are 16/min, and blood pressure is 190/120 mm Hg. The remainder of the examination shows no abnormalities except for an epigastric bruit. Arteriography shows fibromuscular hyperplasia of the left renal artery. The underlying mechanism of this patient's condition was most likely initiated by the release of which of the following substances?

- ☐ A) Aldosterone
- ☐ B) Angiotensin I
- ☐ C) Angiotensin II
- ☐ D) Angiotensin III
- ☐ E) Renin

49. A 47-year-old woman comes to the physician because of right wrist pain that began 2 days ago when she fell on her outstretched arm while playing tennis. She has had limited use of her hand and wrist since the injury occurred. She is otherwise healthy. There is tenderness to palpation over the radial and dorsal aspect of the right anatomic snuffbox. Muscle strength of the right fingers is 5/5, and sensation throughout the hand is intact. Plain x-rays of the hand and wrist show mild degenerative changes at the first carpometacarpal joint. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance that the pain should subside within 1 week
- ☐ B) Arthrography
- ☐ C) Cast immobilization
- ☐ D) Arthroscopy
- ☐ E) Surgical exploration and repair

50. A 9-year-old boy is brought to the physician by his mother because of a 1-month history of intermittent wheezing and mild shortness of breath. The mother reports that his symptoms occur once or twice weekly, occasionally before exercise, and never at night. When his symptoms occur, he must sit and rest. He has no history of serious illness or known allergies. He receives no medications. His older brother and younger sister have asthma. His parents do not smoke cigarettes. The patient's temperature is 37°C (98.6°F), pulse is 92/min, respirations are 16/min, and blood pressure is 90/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. Examination shows enlarged, pale nasal turbinates; mildly enlarged tonsils; and shotty cervical lymphadenopathy. Cardiopulmonary examination shows no abnormalities. Spirometry shows a peak expiratory flow rate of 200 L/min (88% of predicted). In addition to short-acting β_2 -adrenergic agonist therapy, which of the following is the most appropriate pharmacotherapy?

- ☐ A) Leukotriene therapy
- ☐ B) Low-dose corticosteroid and long-acting β_2 -adrenergic agonist therapy
- ☐ C) Low-dose corticosteroid therapy
- ☐ D) Medium-dose corticosteroid therapy
- ☐ E) No additional pharmacotherapy is indicated